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Quizterm 2 Review

Sampling, Modeling (Constant, SLR, Loss), **OLS**, **Gradient Descent**, **Feature Engineering** (OHE), **Cross-Validation**, **Regularization**

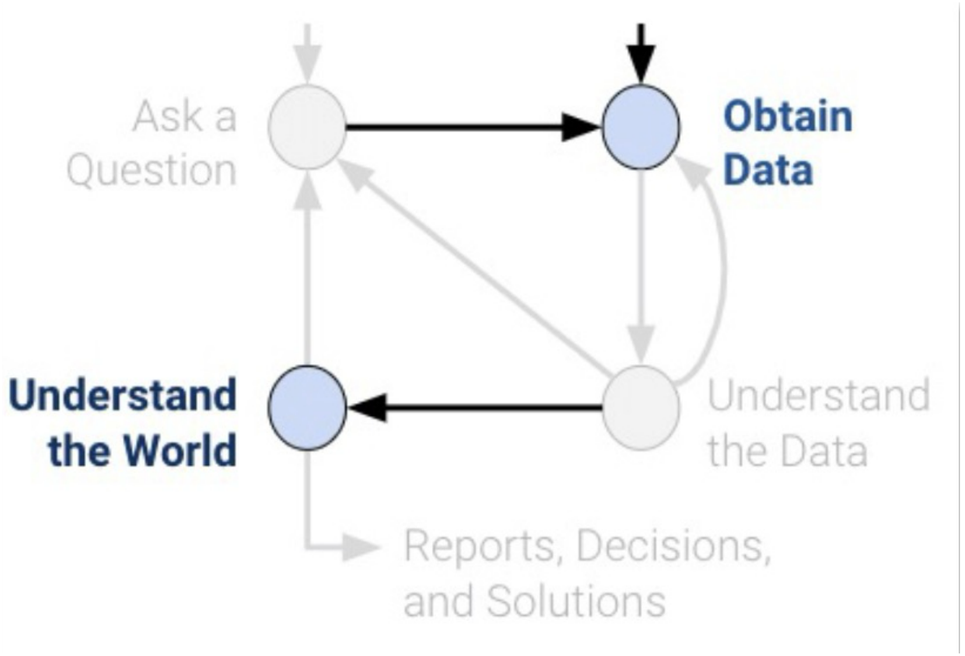
Slides adapted from James Geronimo, Sara Eginova, Sarika Pasumarthy, and Kelly Hu (2025-2026)

Fall 2026

Sampling

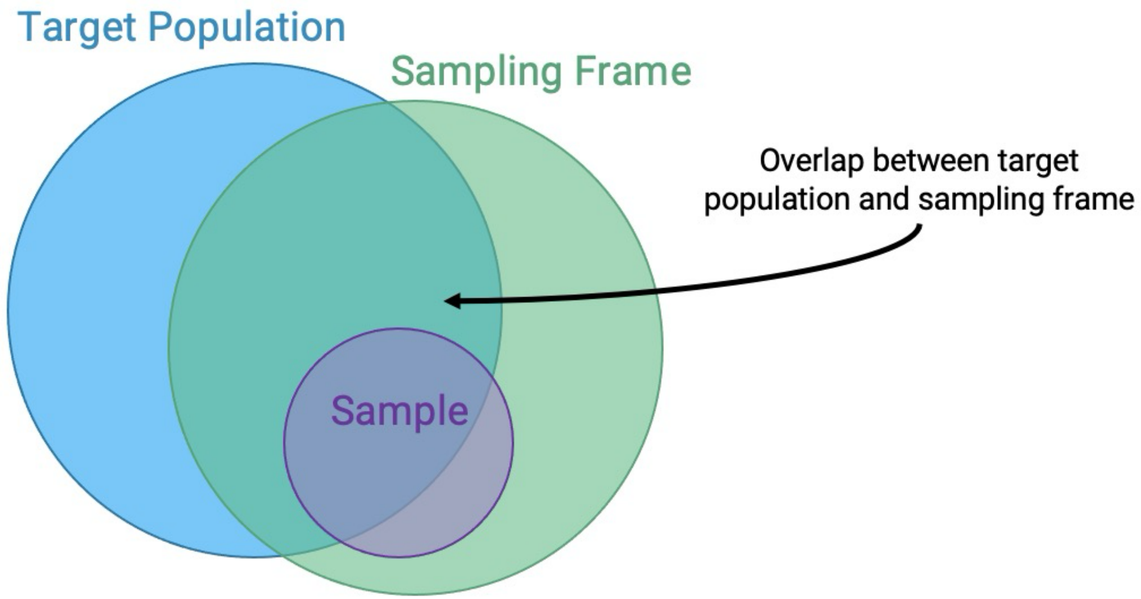
Motivation for Sampling

- Need quality data to understand characteristics of a population
- Extremely difficult to collect all the data describing a population (census)
- **Solution**: Sample *appropriately* from the population



Core Definitions for Sampling

- **Population**: Group you want to learn about
  - **Individuals**: Not always people. (Ex: Bacteria via DNA sequencing, trees, businesses)
- **Sampling Frame**: List from which the sample is drawn
  - Ex: If sampling people, then it's the set of all people that could end up in your sample
- **Sample**: Who you actually end up sampling; Sample is subset of your sampling frame



Sampling Errors

Error	What it means	How to avoid
Selection Bias	Systematically exclude (or favor) particular groups	Random sampling, increase overlap of sampling frame and population
Reporting Bias	Desired measure isn't what is actually measured	Ask better questions with neutral and clear language, ensure anonymity
Non-response Bias	Non-respondents differ from respondents	Increase response rate, follow-up

Other Core Definitions for Sampling

- **Convenience sample**: Whatever you can get a hold of, non-random

- **Probability sample**: Every member in the population has a known probability of being selected
  - Chances don't necessarily be uniform
  - Sample at random based off this known chance

- Randomness benefits:
  - Can measure the errors, estimate bias and chance error, and quantify uncertainty
  - More representative sample of the population, which reduces bias, but only when the probability distribution we're sampling from is accurate

Common Sampling Schemes

- **Uniform random sample with replacement**: is a sample drawn uniformly at random with replacement
  - Note: Random doesn't always mean "uniformly at random"
  - With replacement: Some individuals in the population might get picked more than once

- **Simple random sample (SRS)**: is a sample drawn uniformly at random without replacement
  - Every individual has the same chance of being selected from the sampling frame

- **Stratified random sample**: sampling frame is divided into non-overlapping strata by some chosen characteristics
  - SRS is conducted on each stratum, with sample size proportional to stratum size
  - We can also practice something called **post-stratification** where we weight our data *after* it's been collected so it better matches the known population distribution (HW5Q2e)

Sampling Practice

FA25 MT1 Q5

A Sampling of Errors and Biases

Josh wants to estimate the proportion of UC Berkeley students who prefer cake or pie. He collects a simple random sample (SRS) of **100** emails from the official list of all UC Berkeley students. He emails each selected student and asks them to report whether they prefer cake or pie. **62** students respond to his email within one week. Josh received no additional responses after that.

Jake tells Josh that he could have improved his survey by making two updates:

- Josh should email an SRS of **200** UC Berkeley student emails, instead of **100**.
- Josh should also send a **reminder** email to all students who did not open the initial email within one week.

Assume that Josh could go back in time and make both updates suggested by Jake, and no other changes. Answer the following questions:

Note: You can assume that everyone who fills out either survey will always answer truthfully.

(a) [1 Pt] The chance error of the **updated** survey is _____ the chance error of the **original** survey.

- ◯ Larger than
- ◯ Smaller than
- ◯ The same as

(b) [1 Pt] The ~~response~~ reporting bias of the **updated** survey is likely _____ the response bias of the **original** survey.

- ◯ Larger than
- ◯ Smaller than
- ◯ The same as

FA25 MT1 Q5

A Sampling of Errors and Biases

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- Josh should also send a **reminder** email to all students who did not open the initial email within one week.

Assume that Josh could go back in time and make both updates suggested by Jake, and no other changes. Answer the following questions:

Note: You can assume that everyone who fills out either survey will always answer truthfully.

(c) [1 Pt] The non-response bias of the **updated** survey is likely _____ the non-response bias of the **original** survey.

- ◯ Larger than
- ◯ Smaller than
- ◯ The same as

(d) [1 Pt] The selection bias of the **updated** survey is likely _____ the selection bias of the **original** survey.

- ◯ Larger than
- ◯ Smaller than
- ◯ The same as

FA25 MT1 Q5

A Sampling of Errors and Biases

[SOLUTION]

(a) [1 Pt] The chance error of the **updated** survey is _____ the chance error of the **original** survey.

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Modeling

Modeling

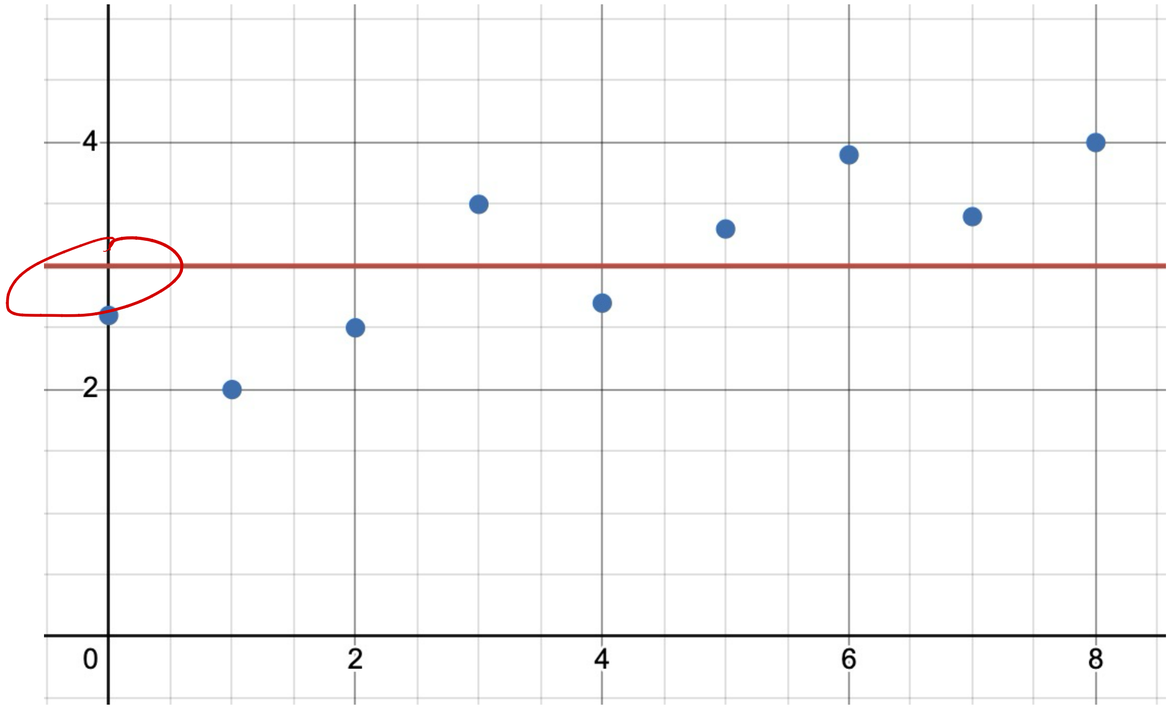
Choose a model	Constant model, SLR, OLS
Choose a loss function	L1 loss (MAE), L2 loss (MSE)
Fit the model	Take derivative of the cost function (average loss) with respect to each model parameter (and set equal to 0)
Evaluate the model	RMSE, Residual plots

Model #1: Constant Model

- AKA summary statistic
- **Always predicts the same constant number**

$$\hat{y} = \theta_0$$

- Ignores any relationships between variables
- Serves as a baseline to compare other complicated models against



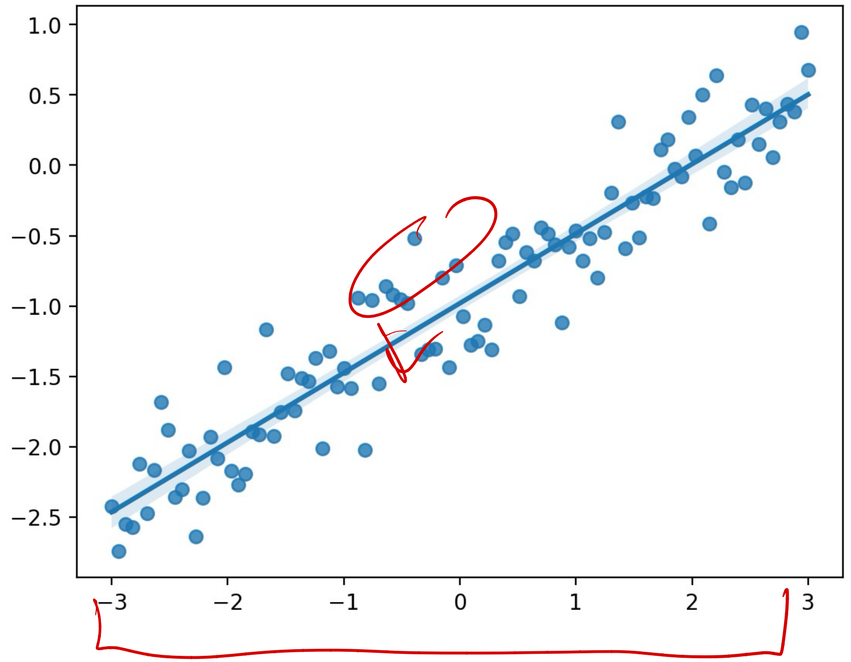
unlike $\hat{y} = \theta_0 + \theta_1 x_1$

no independent variable

Model #2: Simple Linear Regression

$$\hat{y}_i = \theta_0 + \theta_1 x_i$$

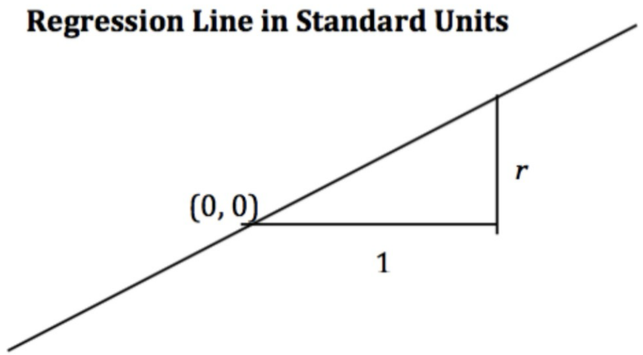
predicted value **parameters** **input/feature**



- **Regression line**:
  - Unique straight line that minimizes the MSE of estimation among all straight lines
  - As with any straight line, defined by a slope and a y-intercept
- Is SLR best fit line always unique? → If there is nonzero variance in the input/feature, yes!

recall: $\hat{\theta}_1 = r\frac{\sigma_y}{\sigma_x}$

Model #2: SLR Model (Best Fit Line)



In standard units:

$$\hat{y}_{su} = r \cdot x_{su}$$

Convert to original units and isolate y:

$$\frac{\hat{y} - \bar{y}}{\sigma_y} = r \cdot \frac{x - \bar{x}}{\sigma_x}$$

$$\hat{y} = \sigma_y \cdot r \cdot \frac{x - \bar{x}}{\sigma_x} + \bar{y}$$

$$\hat{y} = \left(\frac{r\sigma_y}{\sigma_x}\right) \cdot x + \left(\bar{y} - \left(\frac{r\sigma_y}{\sigma_x}\right)\bar{x}\right)$$

Therefore:

$$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \qquad \hat{\theta}_1 = r\frac{\sigma_y}{\sigma_x}$$

Note: These parameters can also be derived by minimizing MSE as we did with the constant model

refsheet!

Fitting the model

Best model → **minimizes loss** (error)

How to minimize loss?

1. Find **derivative** of loss function
2. Find values of theta when derivative is zero
    - If the function is convex, the point with **derivative = 0** is the global minimum

L2 loss + constant model example:

$$\frac{d}{d\theta_0} \mathrm{R}(\theta) = \frac{d}{d\theta_0} \left(\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \right)$$

Use constant model for predicted y:

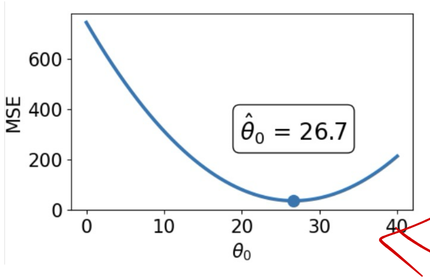
$$\frac{d}{d\theta_0} \mathrm{R}(\theta) = \frac{d}{d\theta_0} \left(\frac{1}{n} \sum_{i=1}^{n} (y_i - \theta_0)^2 \right)$$

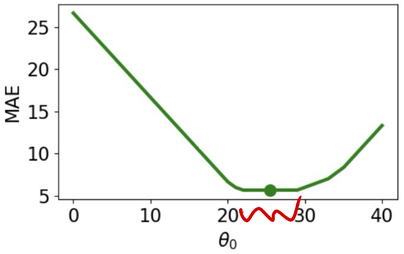
Set derivative to 0 and solve for theta:

$$0 = \frac{-2}{n} \sum_{i=1}^{n} (y_i - \hat{\theta}_0)$$

$$\hat{\theta}_0 = \bar{y}$$

MSE vs. MAE with Constant Model





Mean Squared Error	Mean Absolute Error
Unique solution	Not necessarily unique
Sensitive to outliers	Less sensitive to outliers
Optimal solution = mean(y)	Optimal solution = median(y)
Differentiable	Non-differentiable (piecewise)

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Note: MSE penalizes larger errors more since it takes squares of errors!

Modeling Practice

SU24 MT Q4bc

Lotsa Loss

4 Lotsa Loss [13 Points]

Suppose we have a dataset with n datapoints and one input, $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}$ is the input and $y_i \in \mathbb{R}$ is the target output. We want to use regression on the inputs x_i to predict y_i. Boyu suggests we use the constant model:

$$\hat{y}_i = \theta \text{ where } \theta \in \mathbb{R}.$$

Additionally, he has a new loss function for us, Weighted Mean Squared Error (WMSE). This loss function allows us to up- and down-weight datapoints. For example, perhaps we want to assign higher weights to more recent datapoints. The formula for WMSE is as follows:

$$\text{WMSE} = \frac{1}{n}\sum_{i=1}^{n} w_i(y_i - \hat{y}_i)^2$$

(b) [3 Pts] Take the derivative with respect to θ of the WMSE of the model. **Draw a box around your final answer and be sure to simplify it in terms of** $x_i, y_i, \theta, w_i,$ **and/or** n.

(c) [2 Pts] Now, calculate the value of $\hat{\theta}$, which minimizes the Weighted Mean Squared Error (WMSE) of our model. You may assume, without proof, that the critical point of this loss function is guaranteed to be a minimum. **Draw a box around your final answer and be sure to simplify it in terms of** $x_i, y_i, \theta, w_i,$ **and/or** n.

SU24 MT Q4b

Lotsa Loss

[SOLUTION]

(b) [3 Pts] Take the derivative with respect to θ of the WMSE of the model. **Draw a box around your final answer and be sure to simplify it in terms of $x_i, y_i, \theta, w_i,$ and/or n.**

Solution:

The Weighted Mean Squared Error (WMSE) is given by:

$$\text{WMSE} = \frac{1}{n}\sum_{i=1}^{n} w_i(y_i - \hat{y}_i)^2$$

And our model is the constant model:

$$\hat{y}_i = \theta$$

Thus, the loss function we want to minimize is:

$$L_{\text{WMSE}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} w_i(y_i - \theta)^2$$

$$\begin{aligned}
\frac{\partial L_{\text{WMSE}}}{\partial \theta} &= \frac{\partial}{\partial \theta}\left(\frac{1}{n}\sum_{i=1}^{n} w_i(y_i - \theta)^2\right) \\
&= \frac{1}{n}\sum_{i=1}^{n} \frac{\partial}{\partial \theta}\left[w_i(y_i - \theta)^2\right] && \text{Deriv. of a sum = the sum of the derivatives} \\
&= \frac{1}{n}\sum_{i=1}^{n} w_i \frac{\partial}{\partial \theta}\left[(y_i - \theta)^2\right] && w_i \text{ is a constant with respect to } \theta \\
&= \frac{1}{n}\sum_{i=1}^{n} w_i \cdot 2(y_i - \theta)\frac{\partial}{\partial \theta}(y_i - \theta) && \text{Chain rule} \\
&= \frac{1}{n}\sum_{i=1}^{n} w_i \cdot 2(y_i - \theta)(-1) \\
&= \boxed{-\frac{2}{n}\sum_{i=1}^{n} w_i(y_i - \theta)}
\end{aligned}$$

remember to be careful with
1) chain rule
2) splitting sums

SU24 MT Q4c

Lotsa Loss

[SOLUTION]

(c) [2 Pts] Now, calculate the value of $\hat{\theta}$, which minimizes the Weighted Mean Squared Error (WMSE) of our model. You may assume, without proof, that the critical point of this loss function is guaranteed to be a minimum. **Draw a box around your final answer and be sure to simplify it in terms of $x_i, y_i, \theta, w_i,$ and/or n.**

Solution: To find the value of θ that minimizes the WMSE of our model ($\hat{\theta}$, we set the derivative from (b) equal to zero and solve for θ:

$$-\frac{2}{n}\sum_{i=1}^{n} w_i(y_i - \theta) = 0$$ Set the derivative we calculated in (b) equal to 0.

$$\sum_{i=1}^{n} w_i(y_i - \theta) = 0$$ Divide both sides by $-\frac{2}{n}$.

$$\sum_{i=1}^{n} (w_i y_i - w_i \theta) = 0$$ Multiply the w_i out.

$$\sum_{i=1}^{n} w_i y_i - \sum_{i=1}^{n} \theta w_i = 0$$ Separate into two summations.

$$\sum_{i=1}^{n} w_i y_i - \theta \sum_{i=1}^{n} w_i = 0$$ Pull θ out of second summation as it a constant.

$$\theta \sum_{i=1}^{n} w_i = \sum_{i=1}^{n} w_i y_i$$ Move $\theta \sum_{i=1}^{n} w_i$ to other side (add $\theta \sum_{i=1}^{n} w_i$ to both sides).

$$\theta = \frac{\sum_{i=1}^{n} w_i y_i}{\sum_{i=1}^{n} w_i}$$ Divide both sides by $\sum_{i=1}^{n} w_i$.

$$\boxed{\hat{\theta} = \frac{\sum_{i=1}^{n} w_i y_i}{\sum_{i=1}^{n} w_i}}$$

remember to be careful with
1) chain rule
2) splitting sums

SU25 F Q2ci

Model Enrollees

Extra Practice

(c) [5 Pts] Sammie wants to fit and assess a couple models.

(i) [2 Pts] First, Sammie wants to find the optimal solution for θ_0 with respect to the objective function below.

$$L(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \left(\frac{\theta_0}{4\theta_1} - \theta_1 x_i \right) \right)^2$$

Find the value of θ_0 that minimizes $L(\theta_0, \theta_1)$.

SU25 F Q2ci

Model Enrollees

[SOLUTION]

(i) [2 Pts] First, Sammie wants to find the optimal solution for θ_0 with respect to the objective function below.

$$L(\theta_0, \theta_1) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \left(\frac{\theta_0}{4\theta_1} - \theta_1 x_i\right)\right)^2$$

Find the value of θ_0 that minimizes $L(\theta_0, \theta_1)$.

Solution:

$$\frac{\partial L}{\partial \theta_0} = \frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial \theta_0}\left(y_i - \left(\frac{\theta_0}{4\theta_1} - \theta_1 x_i\right)\right)^2$$

$$= -\frac{2}{4\theta_1 n}\sum_{i=1}^{n}\left(y_i - \frac{\theta_0}{4\theta_1} + \theta_1 x_i\right) \rightarrow \text{set} = 0 \text{ to minimize}$$

$$0 = -\frac{2}{4\hat{\theta_0} n}\sum_{i=1}^{n}\left(y_i - \frac{\hat{\theta_0}}{4\theta_1} + \theta_1 x_i\right)$$

$$0 = \sum_{i=1}^{n}\left(y_i - \frac{\hat{\theta_0}}{4\theta_1} + \theta_1 x_i\right)$$

$$0 = \sum_{i=1}^{n} y_i - \sum_{i=1}^{n}\frac{\hat{\theta_0}}{4\theta_1} + \sum_{i=1}^{n}\theta_1 x_i$$

$$\sum_{i=1}^{n}\frac{\hat{\theta_0}}{4\theta_1} = \sum_{i=1}^{n} y_i + \theta_1 x_i$$

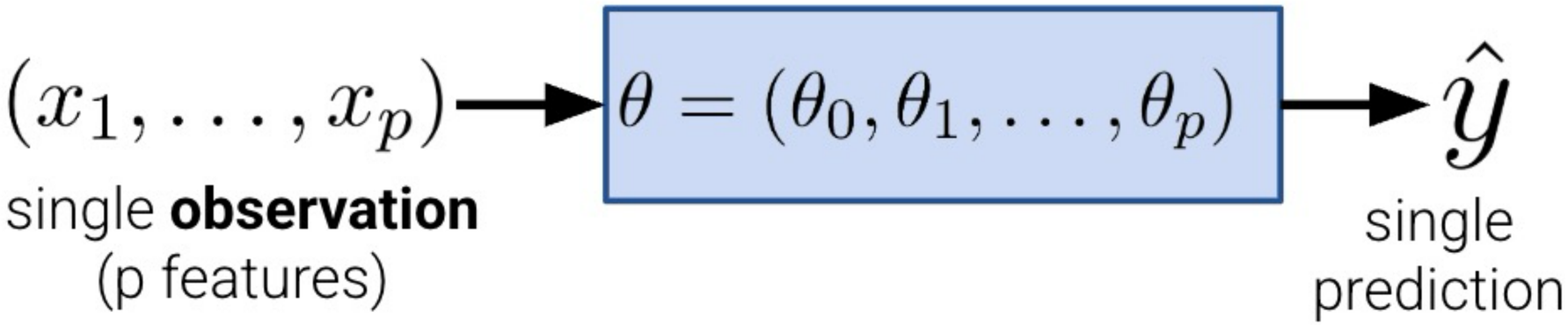
$$\hat{\theta_0} = \frac{4\theta_1}{n}\sum_{i=1}^{n} y_i + \theta_1 x_i$$

OLS

Ordinary Least Squares

Motivation: Multiple Linear Regression i.e. we want to include more than 1 feature!

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_p x_p$$



Matrix Representation

This is a linear model, meaning it is a linear combination of parameters.

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ 1 & x_{31} & x_{32} & \dots & x_{3p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_p \end{bmatrix}$$

$p+1$

n

$$\hat{\mathbb{Y}} = \mathbb{X}\theta$$

Prediction vector	Design matrix	Parameter vector
$\mathbb{R}^n$	$\mathbb{R}^{n \times (p+1)}$	$\mathbb{R}^{(p+1)}$

Note: True output (Y) is also a vector in this case.

Measuring Loss with OLS

MSE metric extended to this new matrix model:

$$R(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$= \frac{1}{n}||\mathbb{Y} - \hat{\mathbb{Y}}||_2^2$$

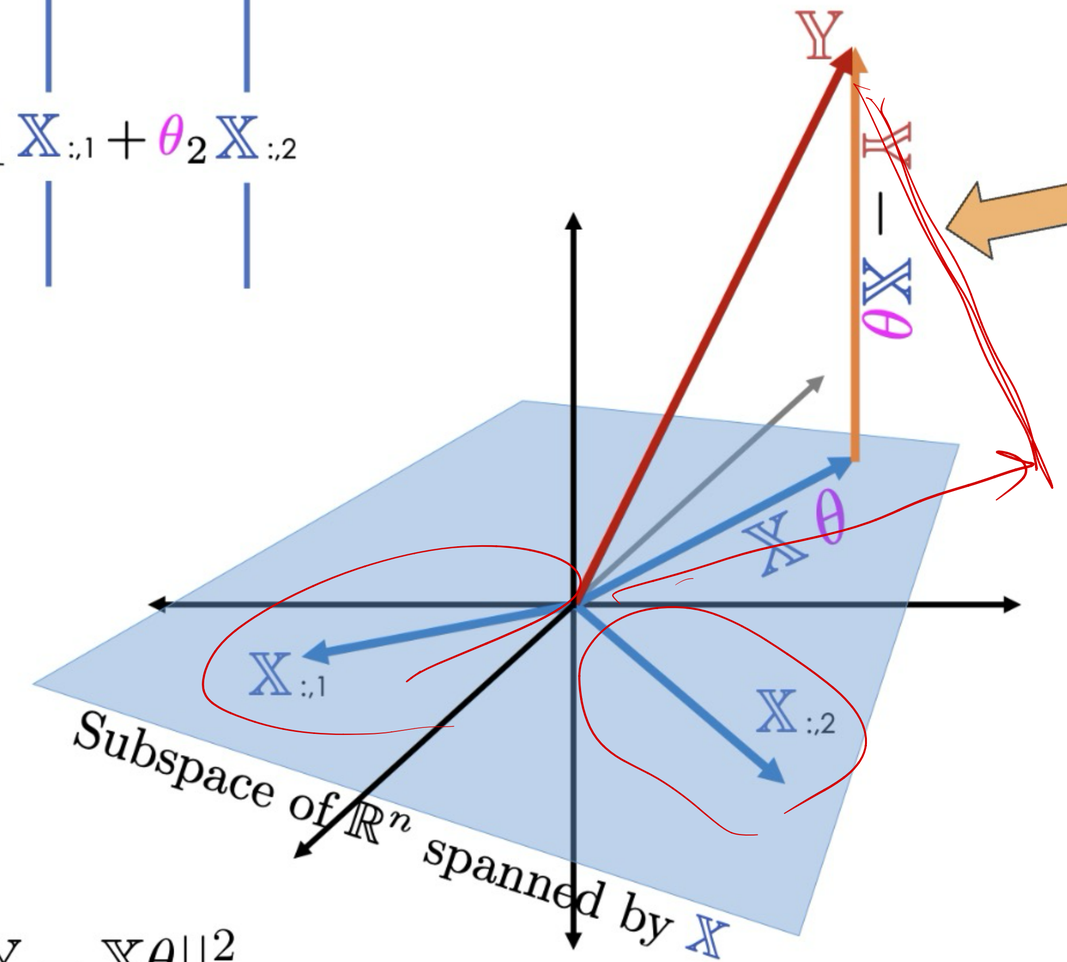
squared L2 norm

Remember that our prediction vector is our design matrix times parameter vector

$$\hat{\mathbb{Y}} = \mathbb{X}\theta$$

Geometric View of OLS

$$\underset{n}{\begin{bmatrix} | \\ \hat{\mathbb{Y}} \\ | \end{bmatrix}}_{1} = \theta_1 \mathbb{X}_{:,1} + \theta_2 \mathbb{X}_{:,2}$$

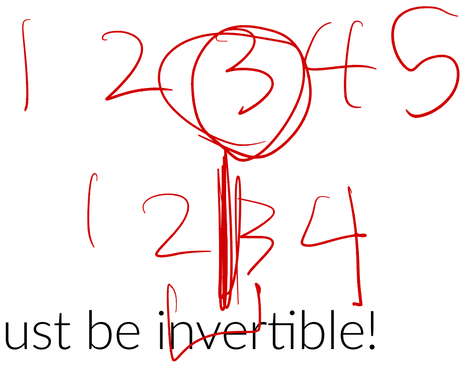


This is the residual vector, $e = \mathbb{Y} - \hat{\mathbb{Y}}$.

Goal:
Minimize the *L2* norm of the residual vector.
i.e., get the predictions $\hat{\mathbb{Y}}$ to be "as close" to our true y values as possible.

$$R(\theta) = \frac{1}{n}||\mathbb{Y} - \mathbb{X}\theta||_2^2$$

The Normal Equation



What is the optimal value of the parameter vector to minimize L2 loss?

$$\hat{\theta} = (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\mathbb{Y}$$ where $\mathbb{X}^T\mathbb{X}$ must be invertible!

	Model	Estimate	Unique?
Constant Model + MSE	$\hat{y} = \theta_0$	$\hat{\theta}_0 = mean(y) = \bar{y}$	**Yes**. Any set of n values has a unique mean.
Constant Model + MAE	$\hat{y} = \theta_0$	$\hat{\theta}_0 = median(y)$	**Yes**, if n is odd. **No**, if n is even. Return average of middle 2 values.
Simple Linear Regression + MSE	$\hat{y} = \theta_0 + \theta_1 x$	$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1\bar{x}$ $\hat{\theta}_1 = r\frac{\sigma_y}{\sigma_x}$	**Yes**. Any set of n (non-constant) values has a unique mean, SD, and correlation coefficient.
Ordinary Least Squares (Linear Model + MSE)	$\hat{\mathbb{Y}} = \mathbb{X}\theta$	$\hat{\theta} = (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\mathbb{Y}$	**Yes**, if $\mathbb{X}$ is **full col rank** (i.e., all cols **linearly independent**)

OLS Practice

SU25 F Q2a

Model Enrollees

2 A model enrollee in Data 100 [16 Pts]

CDSS wants to predict the number of students who will enroll in its courses. The CDSS course coordinators request Data 100 staff to build a variety of linear models to assist their prediction.

(a) [2 Pts] Jake receives enrollment numbers from various past courses. He is interested in creating a constant model ($\hat{y} = \theta_0$) to make predictions. Jake wants to decide between optimizing for MSE vs MAE. Select all statements below that are true.

- ☒ Jake can use gradient descent to find a value of θ_0 that minimizes MAE.
- ☒ It is possible for two different constant predictions to both result in the optimal MAE.
- ☒ Since the MSE loss surface is always convex and smooth, there is always one optimal $\hat{\theta}_0$ that minimizes it.
- ☐ Suppose Jake chooses to optimize the MSE. The median of the data can result in a lower training loss than the training loss associated with the mean.

SU25 F Q2a

Model Enrollees

[SOLUTION]

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- ☐ Suppose Jake chooses to optimize the MSE. The median of the data can result in a lower training loss than the training loss associated with the mean.

SU25 F Q2b

Model Enrollees

(b) [3 Pts] Wesley has $n = 30$ unique data points and 2 features. He fits the following OLS model:

$$\hat{y}_i = \hat{\theta}_0 + \hat{\theta}_1 (x_{i1})^3 + \hat{\theta}_2 \ln x_{i2}$$

Wesley correctly calculates a **unique optimal solution** for his OLS model. Select all statements about Wesley's fitted model (which uses the optimal parameters) that must be true:

- ☐ It is highly likely, but not guaranteed, that the sum of the residuals is 0.
- ☐ The rank of the design matrix ($\mathbb{X}$) must be less than 3.
- ☐ The vector of outcomes ($\mathbb{Y}$) must be in the span of $\mathbb{X}$.
- ☐ The vector of predictions ($\hat{\mathbb{Y}}$) must be a linear combination of the outcomes ($\mathbb{Y}$).
- ☐ If Wesley refit the same model with $(x_1)^6$ as an additional feature, a unique solution will no longer exist.
- ☒ If Wesley refit the same model using L2 (Ridge) regularization and $\lambda = 1$, a unique solution will still exist.

SU25 F Q2b

Model Enrollees

[SOLUTION]

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Wesley correctly calculates a **unique optimal solution** for his OLS model. Select all statements about Wesley's fitted model (which uses the optimal parameters) that must be true:

- ☐ It is highly likely, but not guaranteed, that the sum of the residuals is 0.
- ☐ The rank of the design matrix ($\mathbb{X}$) must be less than 3.
- ☐ The vector of outcomes ($\mathbb{Y}$) must be in the span of $\mathbb{X}$.
- ☐ The vector of predictions ($\hat{\mathbb{Y}}$) must be a linear combination of the outcomes ($\mathbb{Y}$).
- ☐ If Wesley refit the same model with $(x_1)^6$ as an additional feature, a unique solution will no longer exist.
- ☐ **If Wesley refit the same model using L2 (Ridge) regularization and $\lambda = 1$, a unique solution will still exist.**

SU23 MT Q5b

Hat Trick

Extra Practice

	Player	Dis	League	Goals
0	Erling Haaland	55.7	Premier League	1.373626
1	Harry Kane	116.0	Premier League	0.745614
2	Robert Lewandowski	122.6	La Liga	0.945946
3	Marcus Rashford	148.4	Premier League	0.552764
4	Karim Benzema	165.2	La Liga	0.775862

Kippa loves exploring soccer statistics. He obtains a dataset with some of his favorite players' stats from the 2022-2023 season. The dataset contains the following stats for each soccer player:

- `"Dis"`: Total distance, in yards, the player moved the ball, stored as `float`.
- `"League"`: Soccer league of the player's squad, stored as `str`. There are **only two** possible values, La Liga (a Spanish soccer league) or Premier League (an English soccer league).
- `"Goals"`: Average number of goals scored by the player per match, stored as `float`.

Kippa wants to build a model to predict `"Goals"`. He starts with a small subset of five players.

(b) [2 Pts] Kippa uses `"Dis"` and `"League"` as features to build a linear model to predict `"Goals"`. After one-hot-encoding `"League"`, he constructs a design matrix $\mathbb{X}$ and solves the ordinary least squares normal equation to estimate $\hat{\theta}$. He finds that:

- The normal equation $(\mathbb{X}^\top \mathbb{X})\hat{\theta} = \mathbb{X}^\top \mathbb{Y}$ produces a **unique solution** for $\hat{\theta}$
- The residuals $e = \mathbb{Y} - \hat{\mathbb{Y}}$ **sum up to 0**.

Which of the following could **possibly** be his design matrix $\mathbb{X}$? Select all that apply.

☐ A. $\begin{bmatrix} 55.7 & 1 & 0 \\ 116.0 & 1 & 0 \\ 122.6 & 0 & 1 \\ 148.4 & 1 & 0 \\ 165.2 & 0 & 1 \end{bmatrix}$

☐ B. $\begin{bmatrix} 55.7 & 1 \\ 116.0 & 1 \\ 122.6 & 0 \\ 148.4 & 1 \\ 165.2 & 0 \end{bmatrix}$

☐ C. $\begin{bmatrix} 1 & 55.7 & 1 & 0 \\ 1 & 116.0 & 1 & 0 \\ 1 & 122.6 & 0 & 1 \\ 1 & 148.4 & 1 & 0 \\ 1 & 165.2 & 0 & 1 \end{bmatrix}$

☐ D. $\begin{bmatrix} 1 & 55.7 & 1 \\ 1 & 116.0 & 1 \\ 1 & 122.6 & 0 \\ 1 & 148.4 & 1 \\ 1 & 165.2 & 0 \end{bmatrix}$

SU23 MT Q5b

Hat Trick

[SOLUTION]

	Player	Dis	League	Goals
0	Erling Haaland	55.7	Premier League	1.373626
1	Harry Kane	116.0	Premier League	0.745614
2	Robert Lewandowski	122.6	La Liga	0.945946
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Which of the following could **possibly** be his design matrix $\mathbb{X}$? Select all that apply.

☐ **A.** $\begin{bmatrix} 55.7 & 1 & 0 \\ 116.0 & 1 & 0 \\ 122.6 & 0 & 1 \\ 148.4 & 1 & 0 \\ 165.2 & 0 & 1 \end{bmatrix}$

☐ B. $\begin{bmatrix} 55.7 & 1 \\ 116.0 & 1 \\ 122.6 & 0 \\ 148.4 & 1 \\ 165.2 & 0 \end{bmatrix}$

☐ C. $\begin{bmatrix} 1 & 55.7 & 1 & 0 \\ 1 & 116.0 & 1 & 0 \\ 1 & 122.6 & 0 & 1 \\ 1 & 148.4 & 1 & 0 \\ 1 & 165.2 & 0 & 1 \end{bmatrix}$

☐ **D.** $\begin{bmatrix} 1 & 55.7 & 1 \\ 1 & 116.0 & 1 \\ 1 & 122.6 & 0 \\ 1 & 148.4 & 1 \\ 1 & 165.2 & 0 \end{bmatrix}$

Solution: This problem tests the properties of OLS. We know that

1. OLS produces a unique solution $\hat{\theta}$ if $\mathbb{X}$ has full column rank, and
2. The residuals sum up to zero if we include an intercept term.

The first property would rule out C since its first, third, and fourth columns are not linearly independent, making the matrix not full rank.
The second property rules out B since there's no intercept/bias column.
A is correct even without an intercept column because a bias/intercept column of all 1's is in the column space of $\mathbb{X}$. We can see this by directly adding the two one-hot-encoded columns.

Gradient Descent

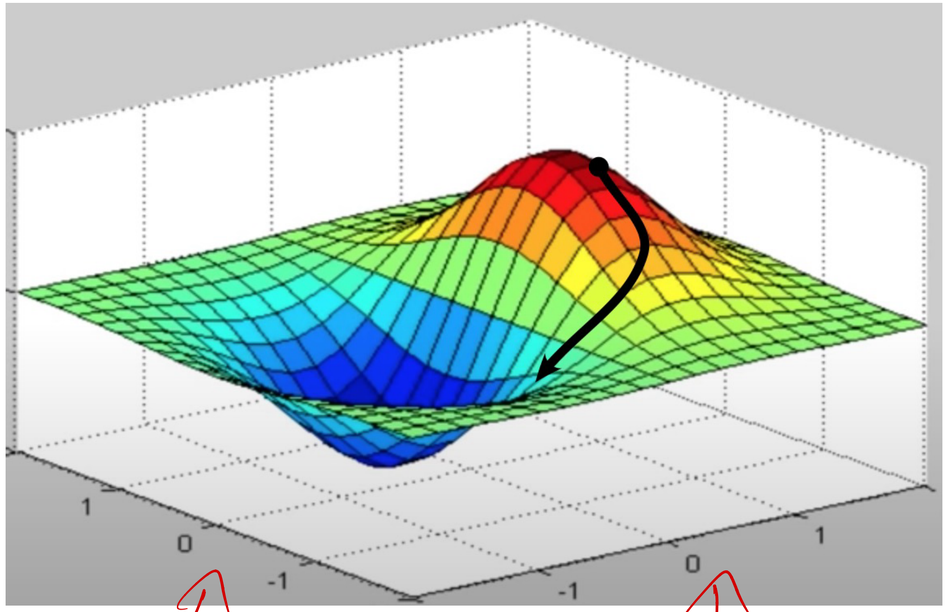
Gradient Descent

$$\hat{Y} = X\hat{\theta}$$

Sometimes, the loss function isn't as simple! Gradient descent is a new optimization technique for more complex loss functions.

Big idea: use an iterative algorithm to numerically compute the minimum of the loss.

MSE ⇒ OLS

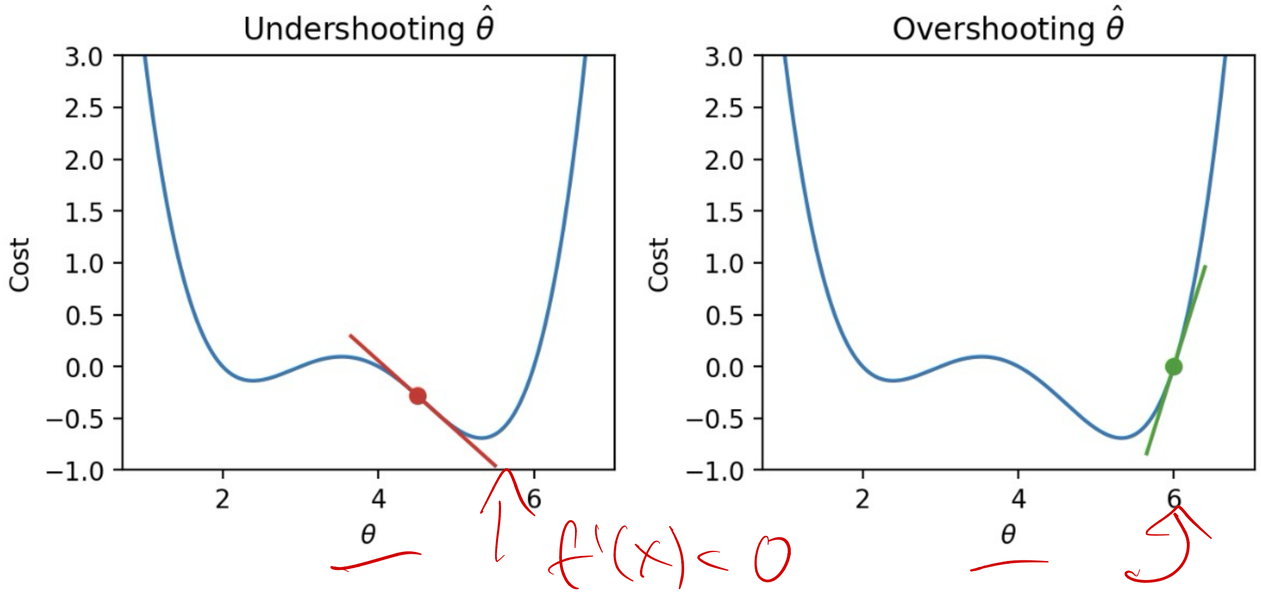


2 params

Can use with other loss functions + models non-linear in params!

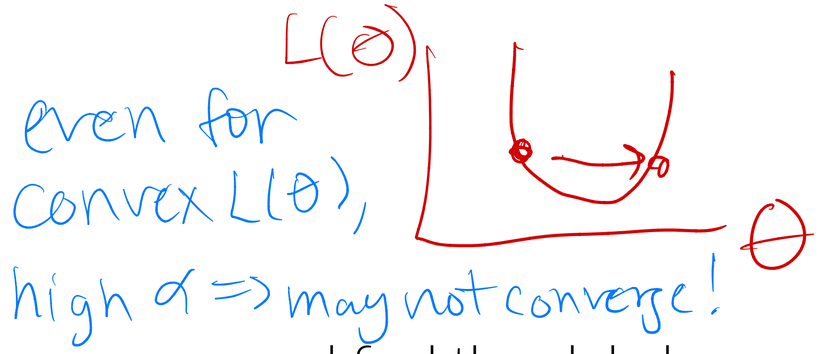
Intuition Behind Gradient Descent

1. Start at an **arbitrary guess.**
2. Use the **first derivative** to determine where to guess next.
    - First derivative is negative --> function decreasing, we can still go lower. Step right to correct.
    - First derivative is positive --> function increasing, we can still go lower. Step left to correct.



1D Case

Gradient Descent Update Formula



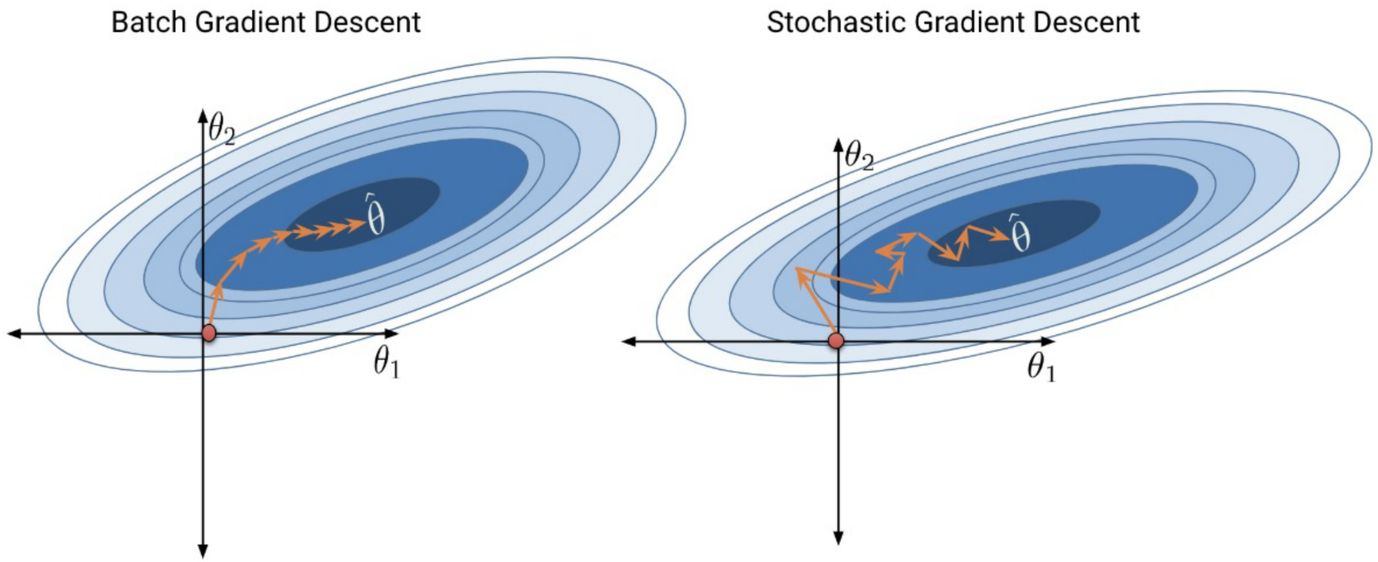
- If the loss function is **convex**, gradient descent is guaranteed to converge and find the global minimum of the objective function with **an appropriate step size and number of iterations**
- GD terminates when the algorithm converges (guesses stop changing)
  - Usually we set number of steps instead of waiting until convergence, because at some point updates are negligibly small with respect to loss improvement
- Alpha (**α**) is the **learning rate** step size
  - It controls the size of the step that we are making!

$$\vec{\theta}^{(t+1)} = \vec{\theta}^{(t)} - \alpha \nabla_{\vec{\theta}} L(\theta^{(t)})$$

opposite

Variants of Gradient Descent

- **Batch GD:** Compute the gradient using the entire dataset
  - Computationally expensive and slow
- **Stochastic (Mini-Batch) GD**: Compute the gradient using a sample of the full dataset
  - Will hop around to converge to the optimal solution because of the element of randomness in which points are used to compute the gradient
- *Epoch*: One full pass through our dataset



Gradient Descent Practice

SU25 F Q2cii

Model Enrollees

(ii) [3 Pts] Sammie chooses a new constant model specification and objective function. She calculates the gradient of her new objective function below with respect to θ_0:

$$L(\theta_0) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \theta_0)^2$$

$$\nabla L(\theta_0) = \left[-\frac{2}{n}\sum_{i=1}^{n}(y_i - \theta_0)\right]$$

Sammie asserts that $\theta_0^{(1)} = 5$, $\alpha = 1$, and $\vec{y} = \begin{bmatrix}1\\2\end{bmatrix}$. ← y_1 ← y_2

Using the values above, calculate $\theta_0^{(0)}$ (i.e., the initializing θ_0 value).

$$5 = \theta_0^{(0)} + 3 - 2\theta_0^{(0)}$$

$$2 = -\theta_0^{(0)}$$

$$\theta_0^{(0)} = -2$$

$$\theta_0^{(1)} = \theta_0^{(0)} - \alpha \nabla L$$

$$\Rightarrow 5 = \theta_0^{(0)} - 1 \cdot \left(-\frac{2}{2}\right)\left((1 - \theta_0^{(0)}) + (2 - \theta_0^{(0)})\right)$$

$$= \theta_0^{(0)} + 1\left(3 - 2\theta_0^{(0)}\right)$$

C

SU23 F Q2cii

Model Enrollees

[SOLUTION]

Sammie asserts that $\theta_0^{(1)} = 5$, $\alpha = 1$, and $\vec{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Using the values above, calculate $\theta_0^{(0)}$ (i.e., the initializing θ_0 value).

Solution:
From the gradient descent update rule:

$$\theta_0^{(1)} = \theta_0^{(0)} - \alpha \nabla L$$

$$5 = \theta_0^{(0)} - 1 * -\frac{2}{2}((1 - \theta_0^{(0)}) + (2 - \theta_0^{(0)}))$$

$$5 = -\theta_0^{(0)} + 3$$

$$-2 = \theta_0^{(0)}$$

SU22 F Q5cd

Slipping Down a New Loss Surface

$$R(\theta) = \frac{1}{2}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

(c) [2 Pts] Using batch gradient descent to minimize $R(\theta)$ from the previous subparts with an arbitrary initialization and learning rate, which of the following is possible? Note that the optimal parameter in this context refers to the parameter which minimizes the empirical risk function on the training set.

- [x] The optimal parameter $\hat{\theta}$ is found in a finite number of gradient steps, and gradient descent returns $\hat{\theta}$.
- [ ] The optimal parameter $\hat{\theta}$ is found in a finite number of gradient steps, but gradient descent does not return $\hat{\theta}$.
- [x] The optimal parameter $\hat{\theta}$ is not found.

α too high?

(d) [2 Pts] Using **stochastic** gradient descent to minimize $R(\theta)$ from the previous subparts with an arbitrary initialization, batch size, and learning rate, which of the following is possible? Note that the optimal parameter in this context refers to the parameter which minimizes the empirical risk function on the training set.

one datapoint

- [x] The optimal parameter $\hat{\theta}$ is found in a finite number of gradient steps, and gradient descent returns $\hat{\theta}$.
- [x] The optimal parameter $\hat{\theta}$ is found in a finite number of gradient steps, but gradient descent does not return $\hat{\theta}$.
- [x] The optimal parameter $\hat{\theta}$ is not found.

noisy update

SU22 F Q5cd

Slipping Down a New Loss Surface

[SOLUTION]

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- ☐ **The optimal parameter $\hat{\theta}$ is not found.**

Feature Engineering

Feature Engineering

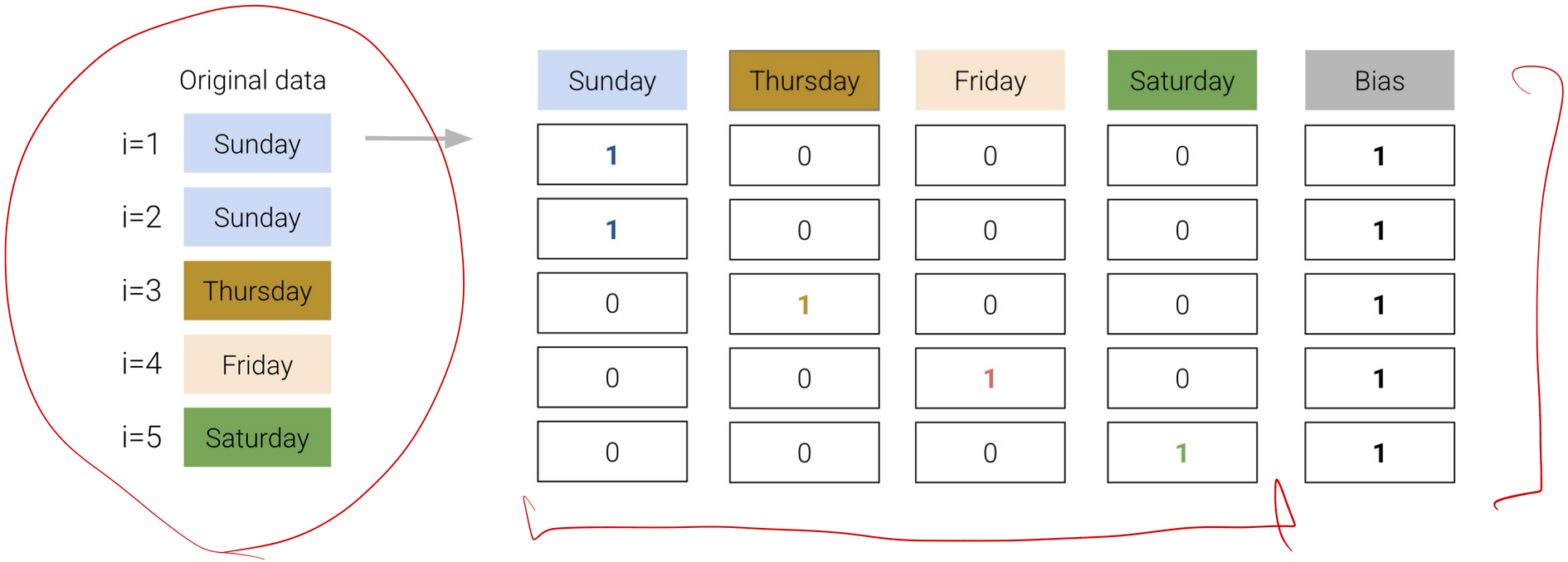
Idea: Transform raw features into more informative features for use in modeling.

Examples:

- linearize features e.g. x => log(x), x => x^3
- transform qualitative (categorical) features (e.g. grade level, flower color) into numerical ones
- create new features from a blend of other features

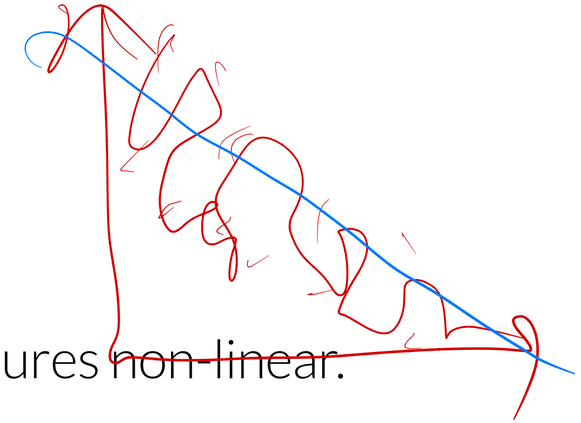
One-Hot Encoding

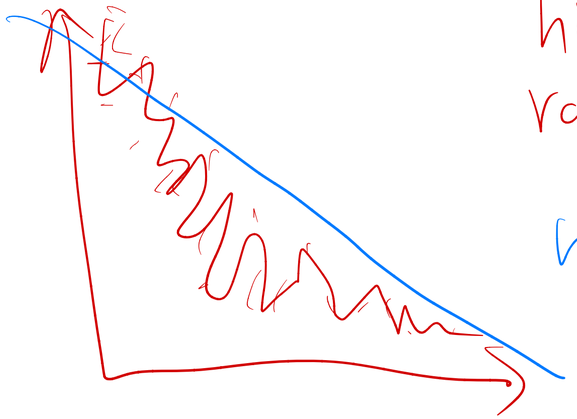
Idea: make column for each possible value in the original qualitative column, and create distinct indicator columns for each value.



Note: If we include a bias column without dropping one of the newly created indicator columns, X is no longer full rank, and we cannot use the normal equation.

Polynomial Features

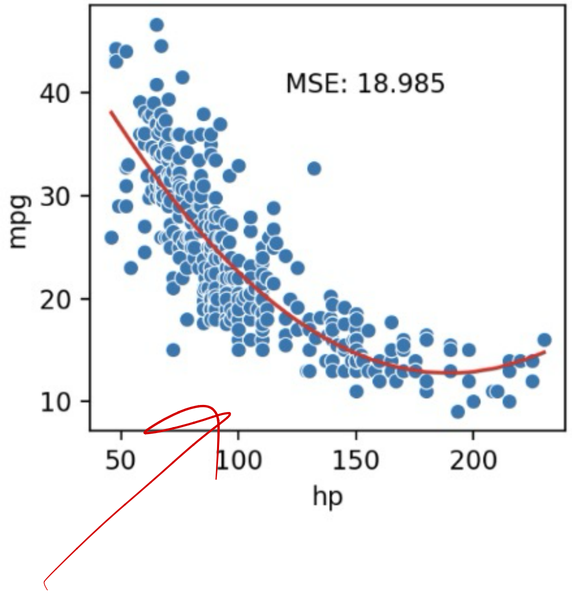




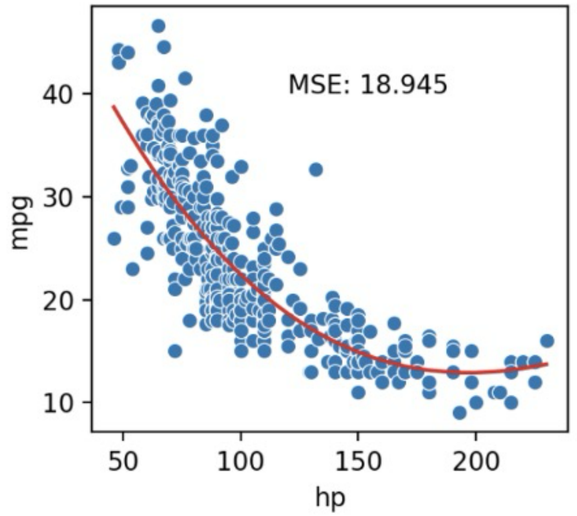
high variance vs. high bias

Idea: Fit linear model better by making features ~~non-linear~~.

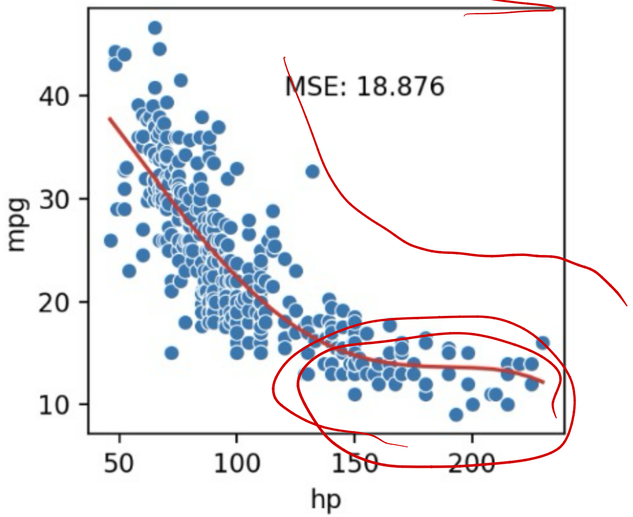
$\hat{y} = \theta_0 + \theta_1(\text{hp}) + \theta_2(\text{hp}^2)$



$\hat{y} = \theta_0 + \theta_1(\text{hp}) + \theta_2(\text{hp}^2) + \theta_3(\text{hp}^3)$



$\hat{y} = \theta_0 + \theta_1(\text{hp}) + \theta_2(\text{hp}^2) + \theta_3(\text{hp}^3) + \theta_4(\text{hp}^4)$



Note: as we add more polynomial features, MSE will *not* increase. But we may **overfit**.

Feature Engineering Practice

FA25 F Q3ci

...and that's a Wrap on Regression

(c) Sarika decides to one-hot encode the `genre` column. When she tries to compute the optimal OLS parameters for her model via the normal equation, she is unable to get a unique solution!

(i) [2 Pts] Which of the following statements **could explain** why this happened?

- ○ True ○ False — The matrix $\mathbb{X}$ does not have full column rank. So, its columns are linearly dependent.
- ○ True ○ False — In the design matrix, Sarika included a bias column and did not remove a reference category from the one-hot encoded columns.
- ○ True ○ False — ~~There is perfect multicollinearity in the features.~~ out of scope
- ○ True ○ False — The dataset has more rows than columns, so the design matrix is not full column rank.
- ○ True ○ False — The matrix $\mathbb{X}^T\mathbb{X}$ is invertible, but the dataset is noisy, so there are many possible OLS solutions.

FA25 F Q3ci

...and that's a Wrap on Regression

[SOLUTION]

(c) Sarika decides to one-hot encode the `genre` column. When she tries to compute the optimal OLS parameters for her model via the normal equation, she is unable to get a unique solution!

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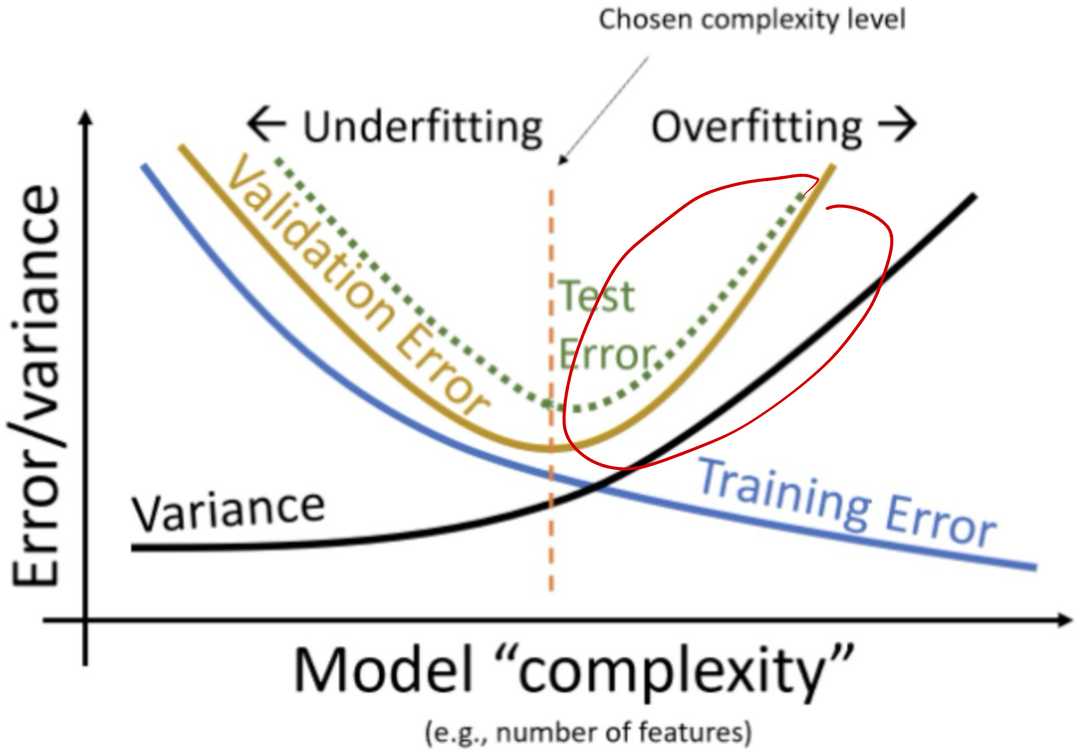
- ◯ **True** ◯ False — There is perfect multicollinearity in the features.
- ◯ True ◯ **False** — The dataset has more rows than columns, so the design matrix is not full column rank.
- ◯ True ◯ **False** — The matrix $\mathbb{X}^T\mathbb{X}$ is invertible, but the dataset is noisy, so there are many possible OLS solutions.

Cross Validation

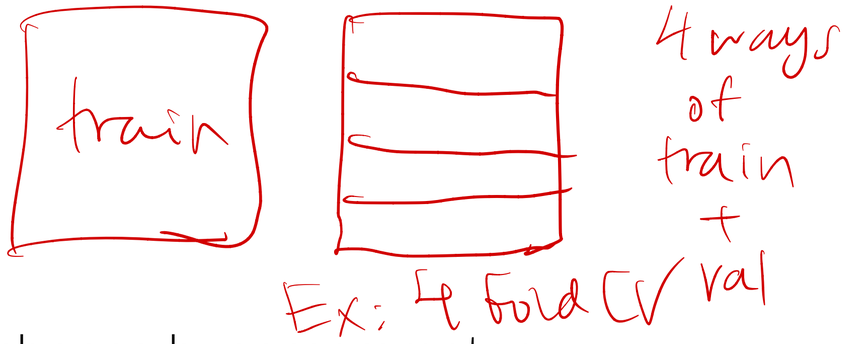
Model Complexity

Key idea: As the model complexity increases, its training error generally decreases.



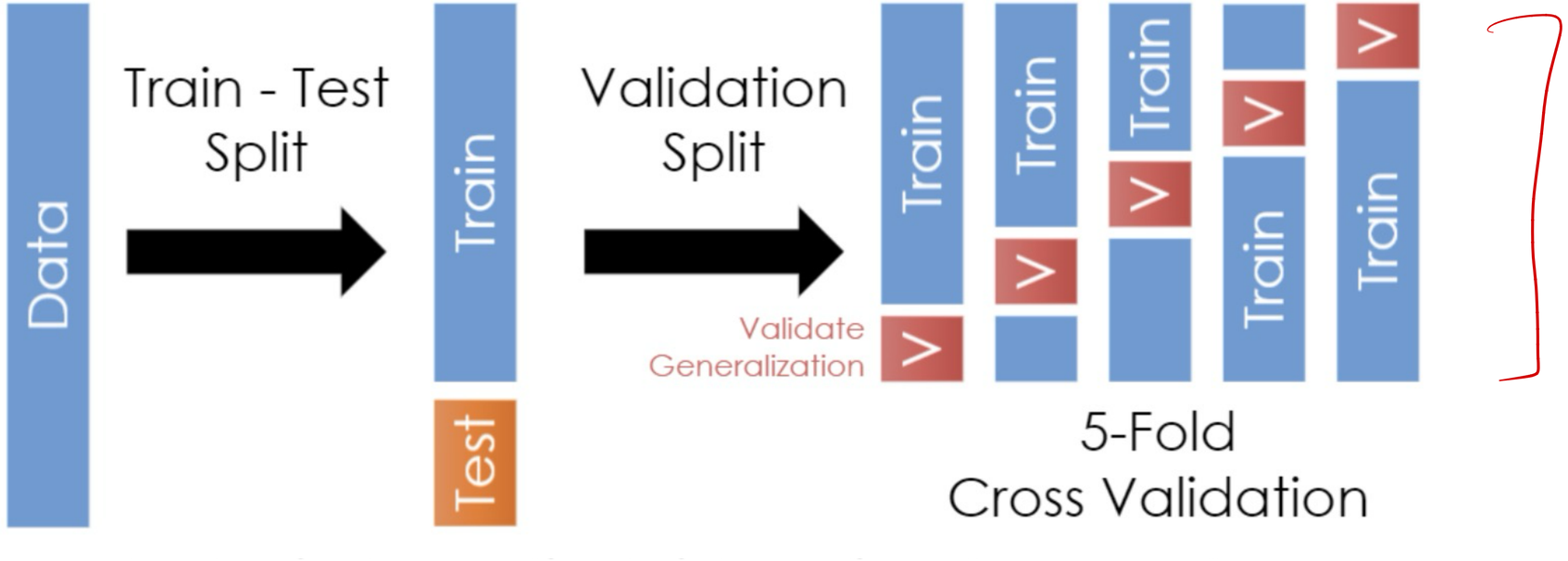
Note: By "model complexity" we generally mean number of features.

Cross Validation (K-Fold)



Idea: Simulate multiple train-validation splits on the training data to choose hyperparameters.

Hyperparameter: a fixed model setting (*e.g. lambda in regularization*, *alpha in gradient descent*) that is configured BEFORE training.



Cross Validation Practice

SU24 MT Q5g

A Sleep Descent

$y = \theta_0 + \theta_1 x$

(g) [2 Pts] You decide to use two hyperparameters, α (the learning rate), and λ (the regularization penalty). You are considering 3 choices of α and 3 choices of λ. If you perform five-fold cross validation to choose the best pair of hyperparameters, how many validation errors would you have to calculate? **Draw a box around your final answer.**

3×3 configs

$3 \times 3 \times 5 = 45$ val errors

SU24 MT Q5g

A Sleep Descent

[SOLUTION]

(g) [2 Pts] You decide to use two hyperparameters, α (the learning rate), and λ (the regularization penalty). You are considering 3 choices of α and 3 choices of λ. If you perform five-fold cross-validation to choose the best pair of hyperparameters, how many validation errors would you have to calculate? **Draw a box around your final answer**.

Solution: With two hyperparameters to tune and three possible choices of each, we have $3 \cdot 3 = 9$ pairs of possible hyperparameters. As we are performing five-fold cross-validation, for each pair of hyperparameters we need to calculate 5 validation errors. Thus, in total we need to calculate validation error $9 \cdot 5 = \boxed{45}$ times.

SU22 F Q4f

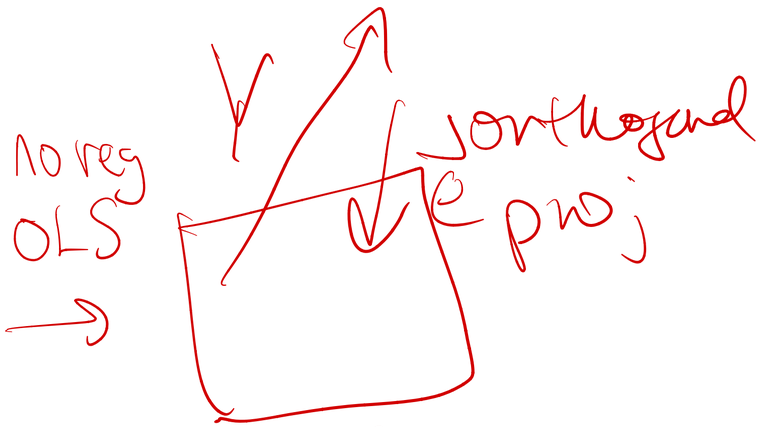
Mmm... More Movie Modeling

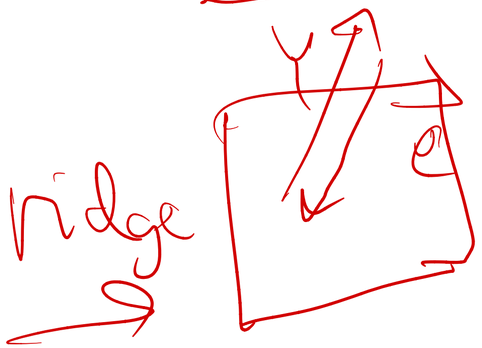
Assume we used the Ridge optimal analytical solution to compute $\hat{\theta}$ in any `fit` call.

reg.

	CV MSE
λ = **0.001000**	70.011580
λ = **0.010000**	62.983541
λ = **0.100000**	151.983646
λ = **1.000000**	80.459269
λ = **10.000000**	139.870794

← avg across all folds





- ☐ $\lambda = 0.01$ is necessarily optimal for minimizing the training mean squared error.
- ☐ $\lambda = 0.01$ is necessarily optimal for minimizing the test mean squared error.
- ☒ For all the models shown, the sum of the residuals is non-zero even if there is a bias term.
- ☒ For all the models shown, the training predictions ($\hat{y} = X\theta$) are within the span of the column vectors of the design matrix X used to train the model.

SU22 F Q4f

Mmm... More Movie Modeling

[SOLUTION]

Assume we used the Ridge optimal analytical solution to compute $\hat{\theta}$ in any `fit` call.

	CV MSE
λ = **0.001000**	70.011580
λ = **0.010000**	62.983541
λ = **0.100000**	151.983646
λ = **1.000000**	80.459269
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- ☐ $\lambda = 0.01$ is necessarily optimal for minimizing the training mean squared error.
- ☐ $\lambda = 0.01$ is necessarily optimal for minimizing the test mean squared error.
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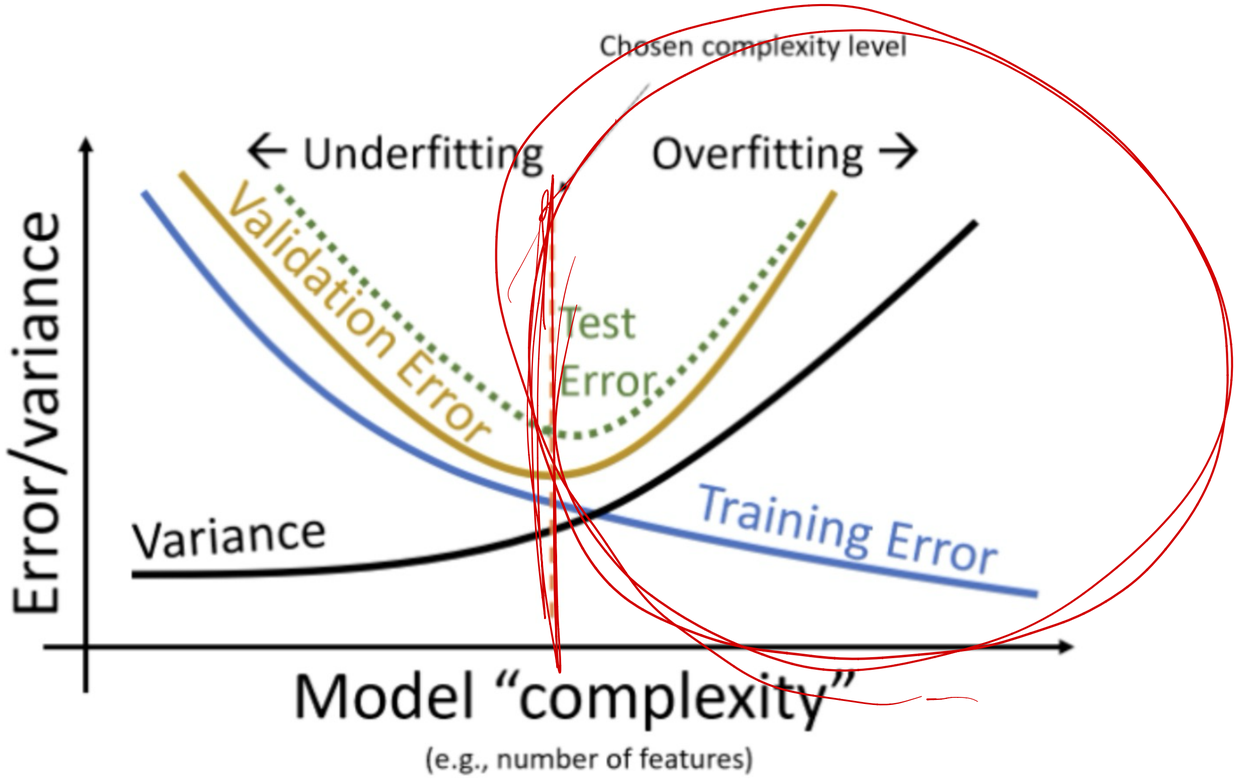
- ☐ **For all the models shown, the training predictions ($\hat{y} = X\theta$) are within the span of the column vectors of the design matrix X used to train the model.**

Regularization

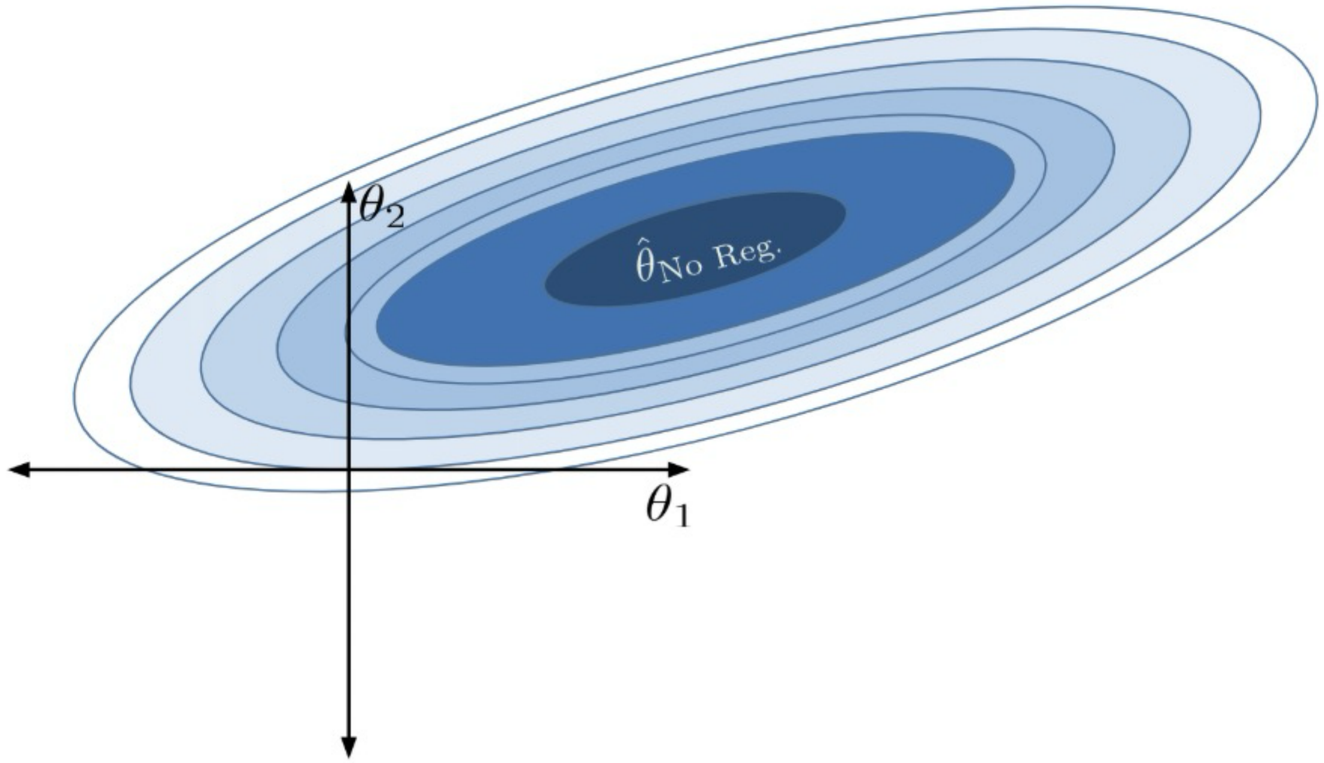
Regularization

Idea: Reduce overfitting by penalizing complex model parameters

- We use too many features --> model complexity and model variance increases

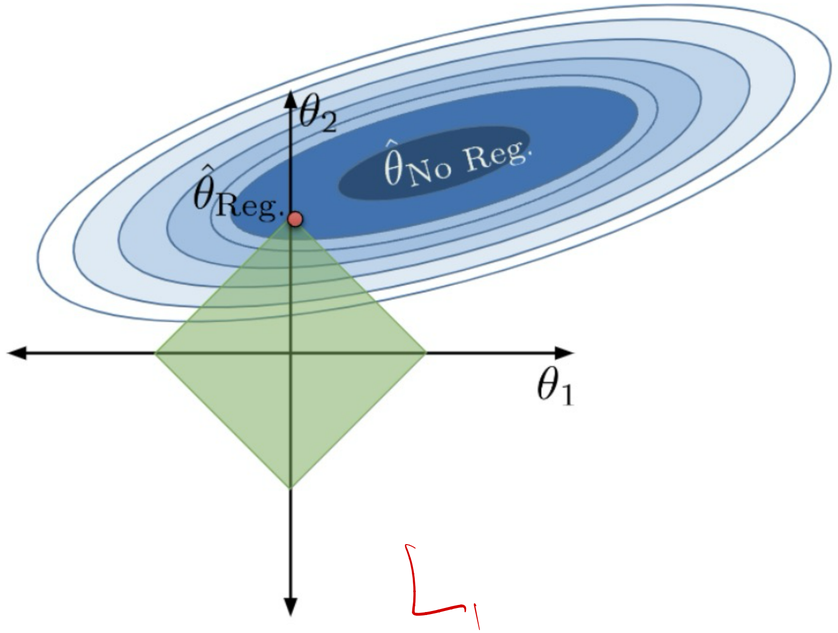


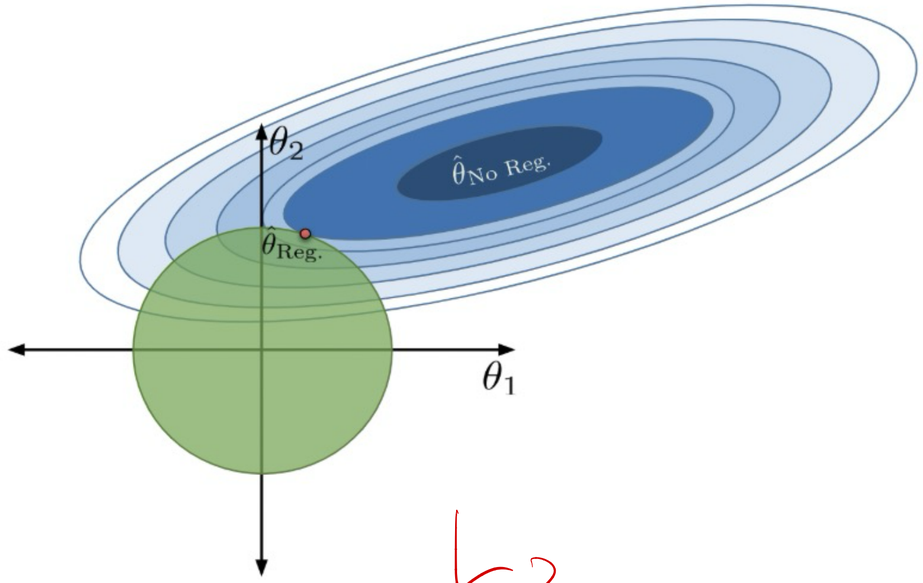
Loss Surface as a Contour Map



Let's Look at the Loss Surface Now!

- The objective with regularization will return the vector that is closest to the optimal vector while staying in the magnitude constraints that we have set
- *This may not be the value that truly minimizes loss*, but that is a tradeoff we accept to have lower model complexity





Regularization Objective

In regression with regularization, we want to minimize the average loss and model complexity.

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \mathbf{Loss}\left(y_i, f_\theta(x_i)\right) + \lambda \mathbf{R}(\theta)$$

Function of model complexity

Regularization Hyperparameter

Note: Larger values of lambda λ = more weight to regularization term = more penalty for higher complexity models.

L1 vs. L2 Regularization Model Complexity

LASSO (L1) Regression:

$$\frac{1}{n}||Y - X\theta||_2^2 + \lambda \sum_{j=1}^{d} |\theta_j|$$

- Encourages sparsity (drives some values of θ to 0)
- No closed-form solution for the optimal θ

"feature selection"

Ridge (L2) Regression:

$$\frac{1}{n}||Y - X\theta||_2^2 + \lambda \sum_{j=1}^{d} \theta_j^2$$

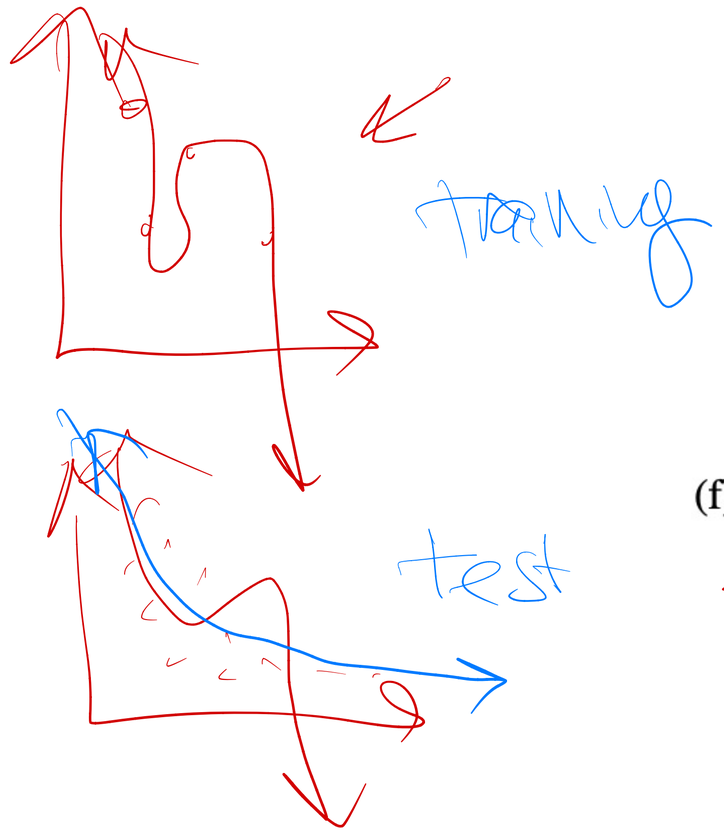
- "Robust" – tends to spread theta weights over many features
- Optimal θ can be found:

$$\hat{\theta} = \left(X^\top X + n\lambda I\right)^{-1} X^T Y$$

Regularization Practice

SU24 MT Q5ef

A Sleep Descent



$(X^TX+n\lambda I)^{-1}X^Ty$

(e) [2 Pts] In the case that the validation and test performance are significantly worse than the training performance, which of the following statements could explain this behavior? **Select all that apply.**

- ☐ The model is underfitting. It might be possible that the model is not expressive enough to model the underlying trend in the data.
- ☐ The model is underfitting. It is possible that there is no underlying trend in our data.
- ☒ The model is overfitting. It is possible that our model is too expressive and started fitting too closely to the noise in our training data.
- ☒ The training data is not representative of the validation and test data, leading to poor performance.
- ☐ None of the above.

(f) [2 Pts] Which of the following statements about regularization are true? Assume we are using OLS. **Select all that apply.**

- ☒ L1 regularization is more likely to lead to a solution in which the coefficients are set to 0 than L2 regularization.
- ☐ L1 regularization is more sensitive to extreme outliers than L2 regularization.
- ☒ L2 regularization has a closed form solution, while L1 regularization does not.
- ☐ Introducing regularization always leads to higher performance on the training set.
- ☐ None of the above.

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[SOLUTION]

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Rohit Vasantha, Zara Wang

Thank you for coming!

Note: random variables, bias-variance tradeoff, parameter inference, SQL, and collinearity are out of scope for Quizterm 2 in Fall 2026 (CCAO is in scope).